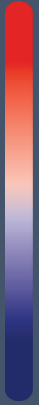


May 2, 2024



ASPEN AEROGELS

Q1 2024 FINANCIAL RESULTS CALL

Disclaimer on Forward Looking Statements

This presentation and any related discussion contains “forward-looking statements” within the meaning of the Private Securities Litigation Reform Act of 1995 that involve risks and uncertainties that could cause actual results to be materially different from historical results or from any future results expressed or implied by such forward-looking statements, including statements relating to Aspen’s 2023 financial outlook. These statements are not historical facts but rather are based on Aspen’s current expectations, estimates and projections regarding Aspen’s business, operations and other factors relating thereto, including with respect to Aspen’s 2024 financial outlook. Words such as “may,” “will,” “could,” “would,” “should,” “anticipate,” “predict,” “potential,” “continue,” “expects,” “intends,” “plans,” “projects,” “believes,” “estimates,” “outlook,” “assumes,” “targets,” “opportunity,” and similar expressions are used to identify these forward-looking statements. Such forward-looking statements include statements regarding, among other things, EV thermal barrier awards and quotes, Aspen’s expectations about capacity, revenue, backlog, costs, expenses, profitability, cash flow, gross profit, gross margin, operating margin, net loss, adjusted EBITDA, adjusted EBITDA margin and related decreases, improvements, timing, variability or trends; beliefs about higher than expected demand from OEM customers and how they may enable path to continued profitability, expectations about improvement in ability to absorb fixed costs and reduction of conversion costs as a percentage of sales and the same leading to target gross margins; beliefs about the general strength, weakness or health of Aspen’s business; acceleration in demand; demand increase from OEM customers, level of penetration in the EV market, and growth in energy industrial markets; beliefs about anticipated battery pack designs from OEMs specifying PyroThin; beliefs about current or future trends in the energy, energy infrastructure, chemical and refinery, LNG, sustainable building materials, EV thermal barrier, EV battery materials or other markets and the impact of these trends on Aspen’s business; beliefs about the strength, effectiveness, productivity, costs, profitability or other fundamentals of Aspen’s business; beliefs about the role of Aspen’s technology and opportunities in the EV market; beliefs about Aspen’s ability to provide and deliver products and services to EV customers; beliefs about content per vehicle, revenue, costs, expenses, profitability, investments or cash flow associated with Aspen’s EV customer opportunities, including the EV thermal barrier business; beliefs about revenue growth and profitability; beliefs about the performance of PyroThin® including its ability to mitigate the propagation of thermal runaway in electric vehicles; beliefs about Aspen’s ability to expand the market for PyroThin, to achieve design wins, to commence shipments of production parts, and to become an industry standard solution for thermal runaway management; beliefs about Aspen’s thermal barrier design, prototype, quoting and assembly activities; expectations about the cost of the capital projects, beliefs about our Mexico assembly facility and its timely operations, and our external manufacturing facility, and their ability to meet the demand; the growth in thermal barrier demand to match the assembly operation and vice versa; and our beliefs about the Department of Energy loan application process. All such forward-looking statements are based on management’s present expectations and are subject to certain factors, risks and uncertainties that may cause actual results, outcome of events, timing and performance to differ materially from those expressed or implied by such statements. These risks and uncertainties include, but are not limited to, the following: inability to execute the growth plan through 2028, inability to manage supply chain disruptions to avoid undue delay or impact on operations, inability to create new product, partnership and market opportunities; any sustained downturn in the industry or energy prices; any sustained downturn in the energy, energy infrastructure, chemical and refinery, LNG, sustainable building materials, EV thermal barrier, EV battery materials or other markets; any failure to sustain project-based demand in the subsea, LNG, on-shore or other markets; the right of EV thermal barrier customers to cancel contracts with Aspen at any time and without penalty; any costs, expenses, or investments incurred by Aspen in excess of projections used to develop pricing under the contracts with EV thermal barrier customers; any failure of Aspen or PyroThin to meet contractual specifications and requirements under contracts with EV thermal barrier customers; Aspen’s inability to create customer or market opportunities for, including PyroThin; any other battery performance and safety products, battery materials or for other new products developed from Aspen’s aerogel technology; any disruption or inability to achieve expected capacity levels in any of our three existing production lines in East Providence, RI or the Mexico assembly facility, or from our third-party suppliers, including in China; any inability to establish or timely establish thermal barrier assembly operations in Mexico or any other location; the failure to receive all regulatory or other approvals required to operate, maintain or expand any of Aspen’s facilities; any failure to achieve demand for Aspen’s products; any failure to achieve expected price increases or average selling prices for Aspen’s products; any significant increase in the cost of raw materials, utilities or any other manufacturing consumable; shortages of raw materials, utilities or any other manufacturing consumable; the failure to generate sufficient operating cash flow or to obtain significant additional capital to pursue Aspen’s strategy; the failure of Aspen’s products to become widely adopted; the competition Aspen faces in its business; any failure to enforce any of Aspen’s patents; any failure to protect or expand Aspen’s Aerogel Technology Platform™; any future finding of invalidity of any of Aspen’s patents in any jurisdiction; any failure to generate sufficient operating cash flow or to obtain sufficient additional capital to continue to pursue Aspen’s new business, technology, patent enforcement, or patent defense strategy; any failure of Aspen’s products to meet applicable specifications and other performance, safety, technical and delivery requirements; the general economic conditions and cyclical demands in the markets that Aspen serves; the economic, operational and political risks associated with sales and expansion of operations in foreign countries, including Mexico; the loss of any direct customer, including distributors, contractors and OEMs; compliance with health and safety laws and regulations; the maintenance and development of distribution channels; and the other risk factors discussed under the heading “Risk Factors” in our Annual Report on Form 10-K for the year ended December 31, 2023 and filed with the Securities and Exchange Commission (“SEC”) on March 7, 2024, as well as any updates to those risk factors filed from time to time in our subsequent periodic and current reports filed with the SEC. All statements contained in this presentation are made only as of the date of this presentation. Aspen does not intend to update this information unless required by law.

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The presentation contains information regarding certain financial targets and outlook with respect to the Company. Such financial targets and outlook constitute forward looking information and is for illustrative purposes only and should not be relied upon as necessarily being indicative of future results. The assumptions and estimates underlying such financial targets are inherently uncertain and are subject to a wide variety of significant business, economic, competitive and other risks and uncertainties, as described above. Actual results may differ materially from the results contemplated in the financial targets contained in this presentation, and the inclusion of such information in this presentation should not be regarded as a representation by any person that the results reflected in such targets will be achieved. "Awards" as used in this presentation is our estimates of the revenues that we believe are potentially derivable from customers based on our knowledge of our customer's selection of our products for use in a specific platform or vehicle lineup. Certain of our customers provide us with various projections of their forecasts for vehicles to be sold in various time frames. Our estimates of Awards are based on this information provided by customers and certain other assumptions, including expectations regarding future sales volumes and the continuation of our relationship with customers. Our customers generally do not guarantee or commit to sales volumes and many of our arrangements with our customers may be terminated by them at any time without penalty.

The information in this presentation is current as of May 2, 2024 and speaks only as of such date. We expressly disclaim any obligation to release any updates or revisions to any information presented herein, including any forward-looking statements, to reflect any change in our expectations or projections or any changes in events, conditions or circumstances on which any such information or statements are based for any reason, except as required by law, even as new information becomes available. All information and forward-looking statements in this presentation are qualified in their entirety by this cautionary statement.

This presentation includes certain non-generally accepted accounting principles (GAAP) financial measures that we use to describe our company's performance or estimated or targeted future performance. The non-GAAP information presented provides investors with additional useful information but should not be considered in isolation or as substitutes for the related GAAP measures. Moreover, other companies may define non-GAAP measures differently, which limits the usefulness of these measures for comparisons with such other companies. We encourage investors to review our financial statements and publicly filed reports in their entirety and not to rely on any single financial measure.

The industry and market data contained in this presentation is based either on our management's own estimates or on independent industry publications, reports by market research firms or other publicized independent sources. Although the Company believes these sources are reliable, it has not independently verified the information and cannot guarantee its accuracy and completeness, as industry and market data are subject to change and cannot always be verified with complete certainty due to limits on the availability and reliability of raw data, the voluntary nature of the data gathering process and other limitations and uncertainties inherent in any statistical survey or market shares. Accordingly, you should be aware that the industry and market data contained in this presentation, and estimates and beliefs based on such data, may not be reliable. Unless otherwise indicated, all information contained in this presentation concerning our industry in general or any segment thereof, including information regarding our general expectations and market opportunity, is based on management's estimates using internal data, data from industry related publications, consumer research and marketing studies and other externally obtained data. Certain financial or other targets are based on management estimates, currently available information and assumptions that may change. Accordingly, there can be no assurance that we will achieve our financial targets or other expectations. The expectations are inherently subject to significant economic, competitive and other uncertainties and contingencies, many of which are beyond the control of management. Actual results may vary materially based on a number of factors. For example, the Company's estimates in this presentation are based on assumptions about the following factors: (i) adoption rates of EV and our customers' market penetration, (ii) the aggregate pricing of our products, and (iii) the average amount of our products incorporated in each vehicle. Due to the rapid pace of technical development in lithium batteries, the Company's assumptions may evolve in the future due to various external factors resulting in different estimates than those arrived here. There are a wide range of estimates being published, including ones that would result in lower per vehicle figures. While the Company's management believes its estimates regarding possible pricing and market size are based on reasonable assumptions and methodology, there can be no assurance that the future pricing and market size will align with the Company's estimates. Such data may be outdated and involves a number of assumptions and limitations and contains projections and estimates of the future performance of the industries in which we operate that are subject to a high degree of uncertainty. We caution you not to give undue weight to such projections, assumptions and estimates.

Q1 2024 Developments

Increasing EV customer demand and continued execution driving strong results and improved 2024 financial outlook

Q1 2024 Highlights¹

- \$94.5M in total revenue, up 107% year-over-year
- Delivered 37% company gross margins
- Quarterly Adjusted EBITDA of \$12.9M (14% margin)²
- Qualified key EI products with external manufacturing facility
- Invoiced PyroThin® parts/prototypes to ~20 programs
- Ended the quarter with \$101.5M in cash & cash equivalents

Near Term Potential Financial Profile

Leveraging existing assets & supply arrangements to target:

\$650M

Annual Revenue
Capacity

\$230M

Annual Potential
Gross Profit

\$160M

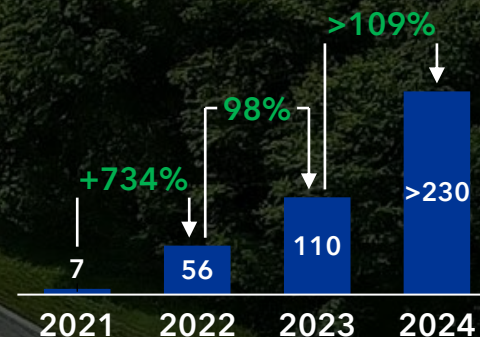
Annual Potential
Adjusted EBITDA

Revenue Trends & 2024 Outlook

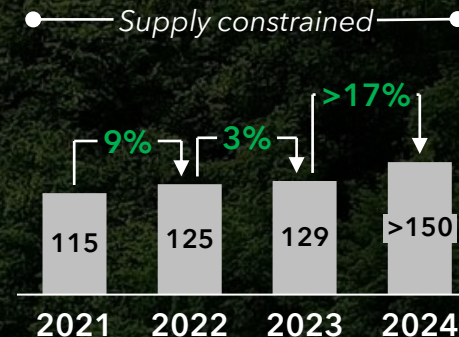
(\$ in millions)



Thermal Barrier



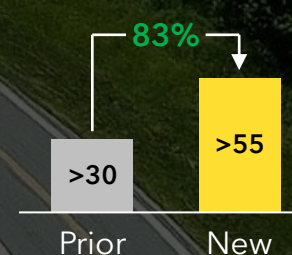
Energy Industrial



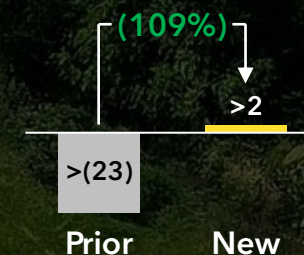
Revenue



Adj. EBITDA²



Net Income (Loss)



Q1 2024 Financial Highlights

Continued driving scalability by managing fixed costs, right-timing CAPEX, and supplying into growing customer demand

\$ in millions except per share figures	Q1 2023	Q1 2024 ¹	LTM Q1 2023	LTM Q1 2024 ¹	Quarterly Commentary
REVENUES <i>YoY Growth</i>	\$45.6 19%	\$94.5 107%	\$187.5 42%	\$287.6 53%	\$65M of EV Thermal Barrier revenues in Q1 led to 5.6x revenue growth year-over-year, headlined by GM's continued ramp \$29M of Energy Industrial revenues in Q1 were supply constrained, on track to deliver \$150M of segment revenues in 2024 2nd Consecutive quarter with >35% gross margin, with Q1 gross profit increasing 6.9x YoY - Over a 10% improvement in material costs as a % of sales - Gross profit improvement and careful OPEX management led to 55% of the \$48.9M in YoY revenue growth falling through to EBITDA - Demonstrated profit potential without relying on outsized revenue growth after having crossed the \$350M annual revenue run rate - Continued investments to deliver on increasing customer demand - Close to breaking even in Q1 2024 \$17.8M Primarily invested in EV Thermal Barrier assembly tooling in Mexico to capture potential H2 2024 and H1 2025 revenue upside - Slowed down Plant II spend to safely secure the site until re-acceleration
Adj. EBITDA² <i>YoY Improvement</i>	\$(13.9) 5%	\$12.9 193%	\$(59.9) (58)%	\$3.9 107%	
Net (Loss) <i>YoY Improvement</i>	\$(16.8) 14%	\$(1.8) 89%	\$(80.0) (59)%	\$(30.9) 61%	
EPS (Loss) <i>YoY Improvement</i>	\$(0.24) 59%	\$(0.02) 92%	\$(1.87) (20)%	\$(0.44) 76%	
Cap Ex <i>YoY Improvement</i>	\$49.4 (241)%	\$25.9 48%	\$212.9 (694)%	\$151.9 29%	

1- Unaudited interim financials for Q1 2024

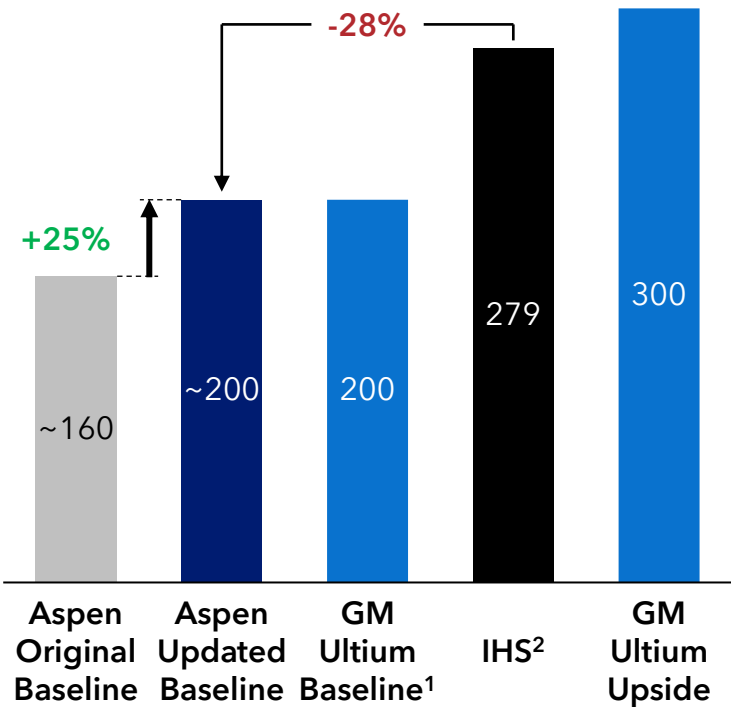
2- See slide 9 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA for the presented period

Updated 2024 Revenue Drivers

Customer volume ramp driving increased thermal barrier baseline revenue of \$230M; Energy Industrial on track for \$150M

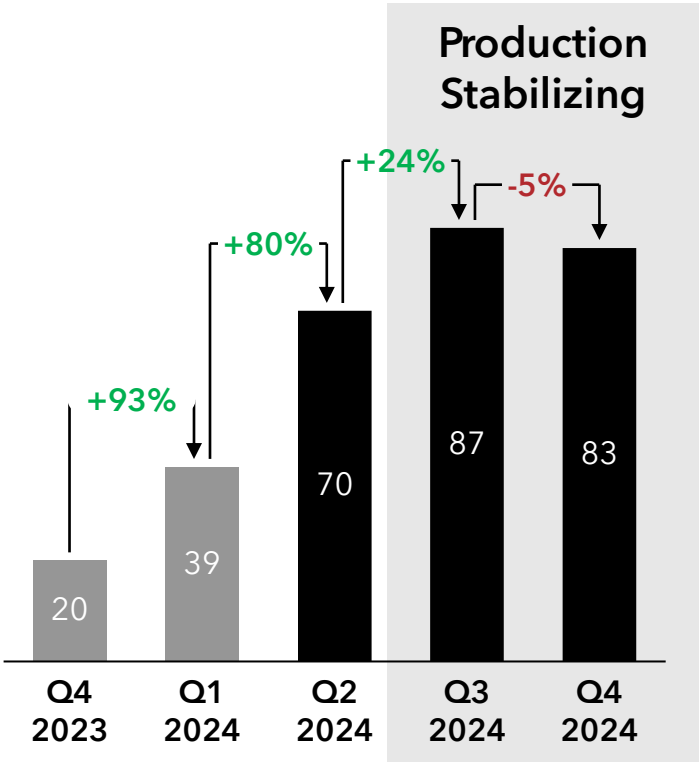
\$230M of Thermal Barrier revenue assumes ~200k Ultium vehicles

Estimated 2024 GM Ultium EV Production
(in thousands of vehicles)



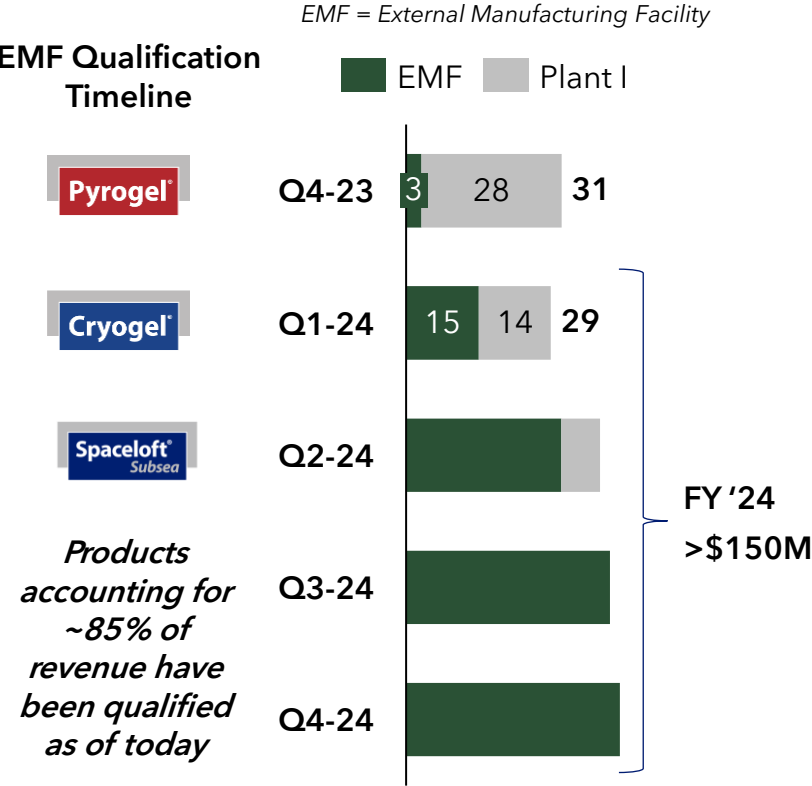
Expecting continued ramp into Q2 & Q3 with potential Q4 tapering

2024 IHS Ultium EV Production Forecast²
(in thousands of vehicles)



Progress with EMF enables \$150M in Energy Industrial revenue outlook

Aspen Energy Industrial Revenue³
(\$ in millions)



1- Per General Motors Q4 2023 earnings call on 1/30/24: "Our plan is to produce and wholesale 200,000 to 300,000 Ultium-based Chevrolet, GMC, Cadillac and BrightDrop EVs in North America this year."
2- IHS April 2024 vehicle production forecast. Please see slide 12 for a further breakdown of volumes
3- Unaudited interim financials for Q1 2024

Updated 2024 Outlook and Historical Trends

Improved 2024 outlook reflects a continuation of plans that have been in place for over 12 months to drive performance

2024 Updated Outlook²

\$ in millions, except per share figures

REVENUES

YoY Growth
Prior

>\$380

>59%
>\$350

Adj. EBITDA²

Prior

>\$55

>\$30

Net Income (Loss)

Prior

>\$2

>\$(23)

EPS (Loss)

Prior

>\$0.03

>\$(0.30)

Cap Ex (excl. Plant II)
+ Plant II Capex (1H / 2H)

\$50

\$30 / \$15*

1- Unaudited interim financials for Q1 2024

2- See slide 9 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA for the presented period

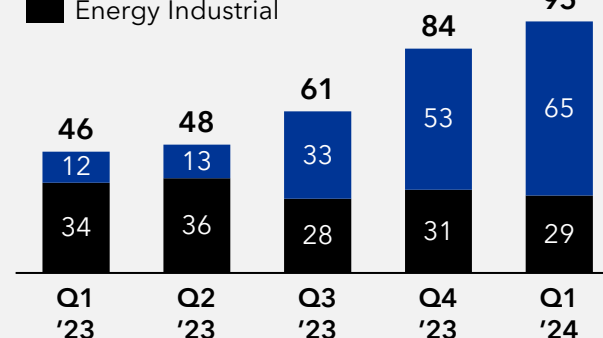
* Assuming continued construction right-timing

15 Month Quarterly Trajectory¹

Revenues

\$ in millions

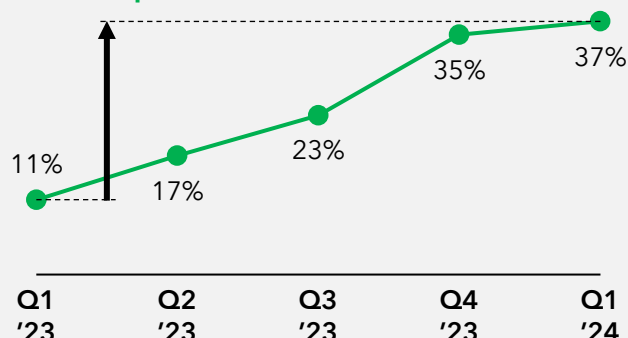
■ Thermal Barrier
■ Energy Industrial



Gross Profit Margin

%

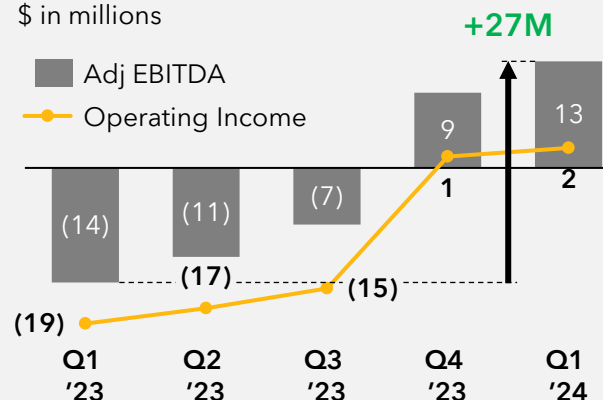
+26% pts



Adj. EBITDA² and Operating Income

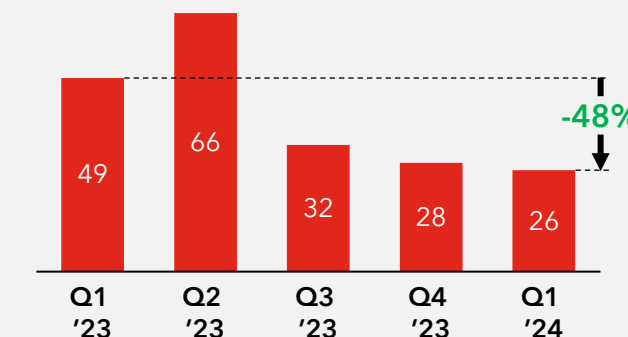
\$ in millions

■ Adj EBITDA
— Operating Income



Capital Expenditures

\$ in millions



Note: Figures may not total due to rounding

Summary

Earning our right to win by leveraging the Aerogel Technology Platform®, executing, and investing to grow profitably

DEMONSTRATED OPERATING EXECUTION

Continued cost discipline and operating efficiencies resulting in increased Adj. EBITDA outlook by 83% to \$55M on \$380M revenue²

INCREASING FLEXIBILITY TO MEET DEMAND

Qualified key products and delivered \$14.6M of Q1 Energy Industrial revenue through external manufacturing facility

ADVANCED PYROTHIN® COMMERCIAL PROGRESS

Continued traction towards diversified customer base; robust pipeline with over \$9B in potential value over the next 5 years

ACCELERATING NEAR-TERM PROFITABILITY

\$94.5 million in Q1 2024 revenue enabled record quarterly gross margins of 37% and Adj. EBITDA of \$12.9M^{1,2}



Appendix

GAAP to Non-GAAP Reconciliation

	2024	2023				LTM		FY 2024 Outlook	
(\$ in millions)	Q1	Q1	Q2	Q3	Q4	Q1 2024	Q1 2023	Prior	Updated
Net Loss	(1.8)	(16.8)	(15.4)	(13.1)	(0.5)	(30.9)	(80.1)	(23.0)	2.0
Depreciation and Amortization	5.8	2.7	3.5	4.6	4.6	18.4	9.8	30.0	30.0
Stock-based Compensation	4.7	2.3	2.7	2.8	3.2	13.4	9.8	14.0	14.0
Other (Income) Expense & Income Tax Expense	4.3	(2.1)	(1.6)	(1.6)	1.9	3.0	0.5	9.0	9.0
Adjusted EBITDA	12.9	(13.9)	(10.8)	(7.3)	9.1	3.9	(59.9)	30.0	55.0

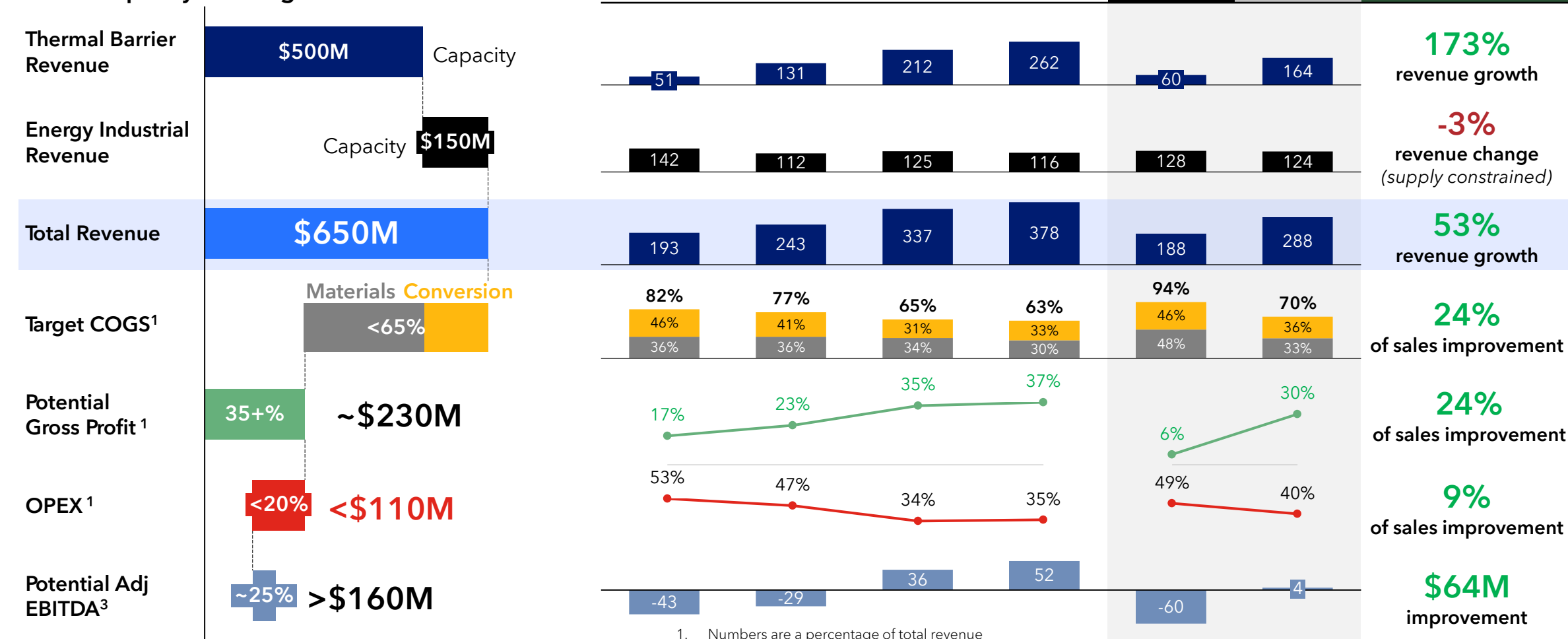
Note: Figures may not total due to rounding

Operating Plan Gearing and Recent Results

Continued progress towards sustainably being a \$650M+ revenue business of 25% EBITDA margins with current capacity

\$ in millions, expressed as *Annual Run Rate*²

Annual Capacity and Target Business Model

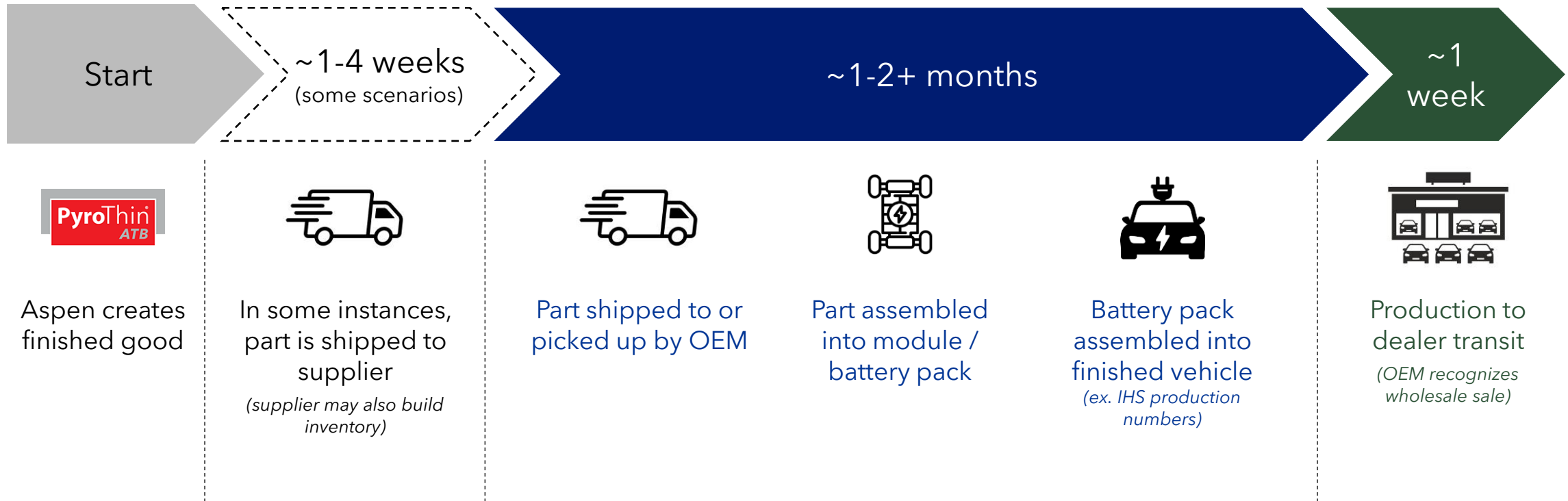


- Numbers are a percentage of total revenue
 - Unaudited interim financials for Q1 2024
 - See slide 9 herein for a reconciliation of net loss, the most directly comparable GAAP measure, to Adjusted EBITDA
- Note: Figures may not total due to rounding

EV Thermal Barrier to Produced Vehicle Timeline

Aspen's finished PyroThin parts can take anywhere from ~1-3+ months to be installed into a produced vehicle

Estimated timeline for average North American PyroThin finished part

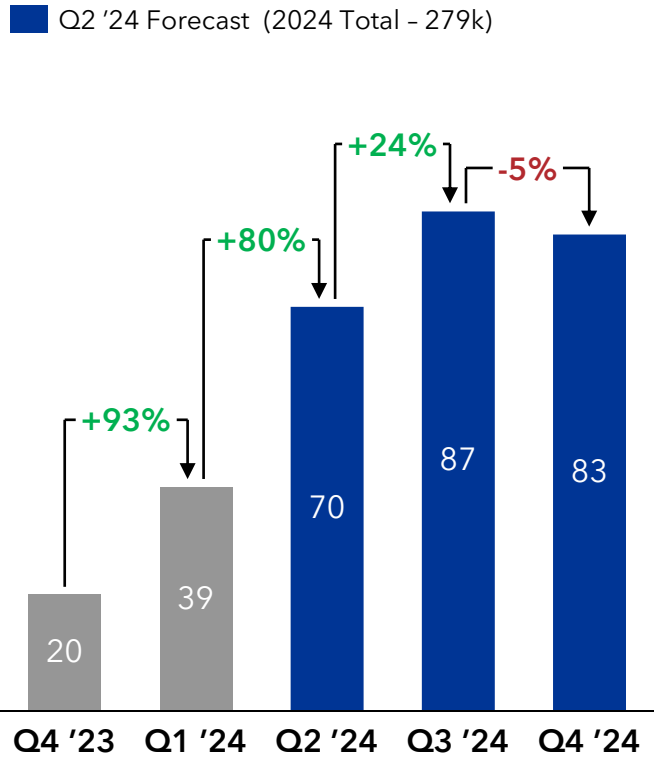


Demand Overview - GM Ultium IHS Forecast

GM Ultium production is expected by IHS to continue increasing over the next two quarters; 2025 / 2026 outlook is strong

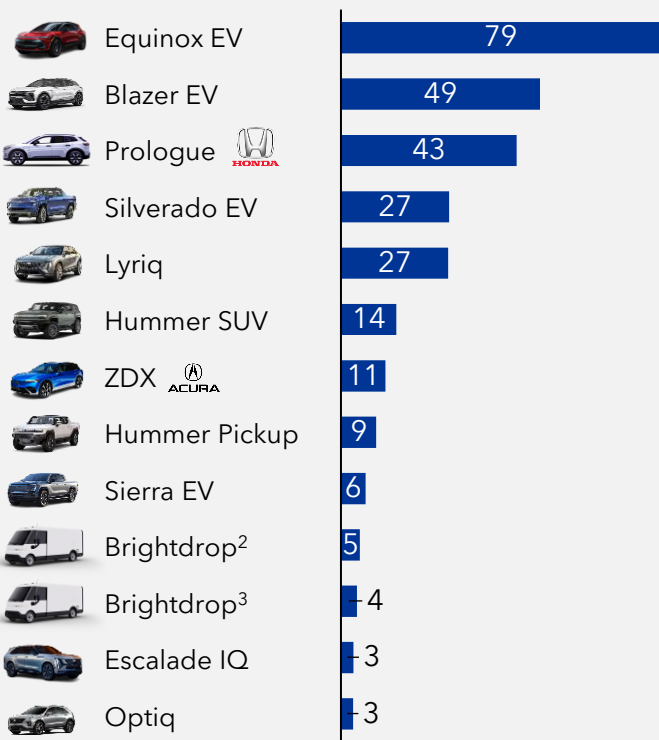
Ultium platform volumes have begun to accelerate

GM Ultium Production Actuals & Forecast¹
(Vehicles in thousands)



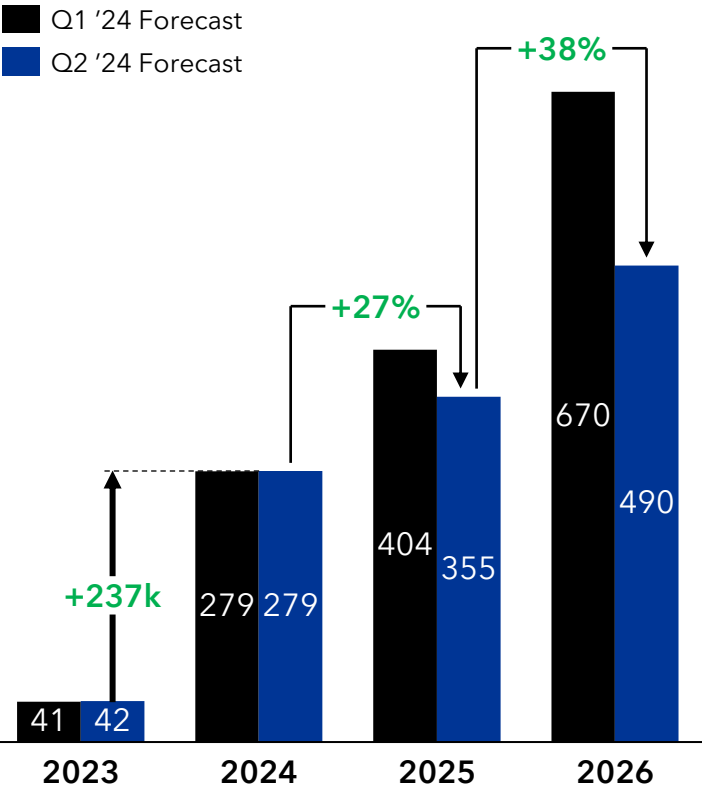
Equinox, Blazer, and Prologue nameplates are ~61% of '24 volume

2024 GM Ultium EV Production Forecast¹
(Vehicles in thousands)



The long-term outlook remains strong

GM Ultium Production Actuals & Forecast⁴
(Vehicles in thousands)

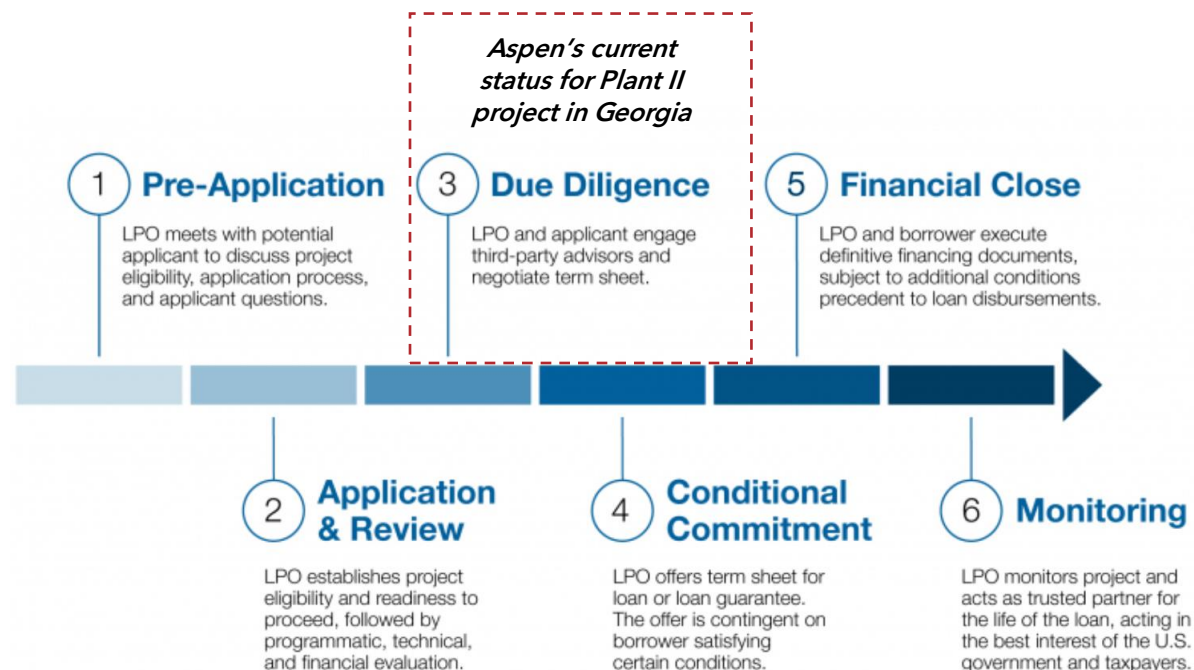


1- IHS April 2024 vehicle production forecast
2. Brightdrop Zevo 600
3. Brightdrop Zevo 400
4. IHS January 2024, April 2024 vehicle production forecasts - 2025 includes reintroduced Chevy Bolt

DOE LPO ATVM Loan Program Overview

Overview

OFFICE: Loan Programs Office	NEW PROGRAM: No, but includes substantive modifications to existing program
FUNDING AMOUNT: \$3,000,000,000 of credit subsidy appropriations under IRA (resulting in an estimated \$40,000,000,000 of direct loans)	FUNDING MECHANISM: Direct loans
RECIPIENTS: A manufacturer of eligible advanced technology vehicles or of qualifying components	PERIOD OF AVAILABILITY: To remain available for commitment through 9/30/2028
ASSISTANCE LISTING: TBD	FORMULA FUNDING: No
TRIBAL ELIGIBILITY: Yes	COST SHARE REQUIREMENT: N/A. Loan amount cannot exceed 80% of eligible project costs.



For further information, please visit the official Department of Energy website

Source: Official DOE & LPO website (<https://www.energy.gov/lpo/application-process>) (<https://www.energy.gov/lpo/advanced-technology-vehicles-manufacturing-loan-program-0>)

DOE LPO ATVM Loan Program Stage Description

Advanced Due Diligence & Term Sheet Negotiation Phase Overview

- LPO performs financial, credit, legal, environmental, and market due diligence.
- LPO typically engages external advisors (e.g., legal, technical, financial, market, insurance) to assist in due diligence and structuring. External advisory costs are borne by the applicant.
- LPO reviews project compliance with federal environmental laws.
- LPO structures the transaction, underwrites and negotiates term sheet with the applicant.
- LPO presents the project for credit approval to internal Project Review Committee, external Interagency Review and internal Credit Review Board.
- LPO and applicant co-sign term sheet.
- LPO issues Conditional Commitment.

For further information, please visit the official Department of Energy website
Source: Official DOE & LPO website (<https://www.energy.gov/lpo/advanced-transportation-financing>)

PyroThin® Thermal Barrier Opportunity Drivers

Eight different macro parameters drive our content per vehicle (CPV) opportunity, and long-term revenue potential

		Commentary
1	Vehicle sales by region North America Europe China Rest of World	- China expected to lead global EV penetration - Higher price and content vehicles in NA and EU
2	Vehicle classification Light-duty (PV and LCV) On-road Heavy Commercial Vehicles Off-road	- Demonstrated commercial success within Light-duty - Most recent award in M/HDT segment highlights commercial vehicle opportunity potential
3	Powertrain technology BEVs PHEV HEV MHEV FCEV ICE	- TP risk increases as OEMs pack more energy density into EV batteries, both in EV's and Plug-In Hybrids - OEMs seeking to provide more range in PHEVs
4	Cell chemistry LFP LMFP Ni-low/mid Ni-rich Semi solid-state	- All battery chemistries have thermal propagation risk - Highest CPV in nickel-based chemistries due to higher energy density
5	Cell form factor Prismatic Pouch Cylindrical	- Pouch and Prismatic expected to continue to account for ~70% of global market - Cylindrical less suited for cell-to-cell solutions
6	Battery capacity PC BEV 50 - 95 kWh PC PHEV 15 - 30 kWh LCV BEV 40 - 145 kWh LCV PHEV 15 - 65 kWh M/HDT BEV 150 - 400 kWh M/HDT PHEV ~100 kWh	- OEMs seeking to increase range, while higher density packs have more risk of thermal propagation - NA and EU vehicles have larger batteries
7	Use case Cell-to-cell Module level Pack level Battery thermal barrier Cabin insulation Body noise dampening	- Aerogel solutions excel as a cell-to-cell barrier - Core requirements are thermal isolation, fire protection and mechanical properties
8	TR/TP mitigation goals No TP Goals Delay TP Stop TP	- Recalls and global regulation expected to continue to driving OEM awareness and Stop TP goals - OEM goals to Stop TP results in more content

Core Focus Area
 Potential Focus Area
 Currently Out of Scope

